

Problem, Chapter-8: Two hundred Engineers were interviewed and classified according to their results and job satisfaction. The distribution of graduates by results and job satisfaction are given in the following contingency table:

Results	Job satisfaction	
	Yes	No
Excellent	20	70
Good	45	65

Test at 5% level of significance whether there is any relationship between results and job satisfaction.

(Note: At 5% level of significance tabulated value of Chi-square @ 1df =3.84)

Solution: We consider the following hypothesis

H0: There is no relationship between results and job satisfaction.

H1: There is a relationship between results and job satisfaction.

Results	Job satisfaction		Total
	Yes	No	
Excellent	20 (O ₁₁)	70(O ₁₂)	90
Good	45(O ₂₁)	65(O ₂₂)	110
Total	65	135	200

We know,
$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Here, $E_{11} = \frac{90 \times 65}{200} = 29.25$, $E_{12} = \frac{90 \times 135}{200} = 60.75$, $E_{21} = \frac{110 \times 65}{200} = 35.75$,

$E_{22} = \frac{110 \times 135}{200} = 74.25$

Therefore,
$$\chi^2 = \frac{(20 - 29.25)^2}{29.25} + \frac{(70 - 60.75)^2}{60.75} + \frac{(45 - 35.75)^2}{35.75} + \frac{(65 - 74.25)^2}{74.25}$$

$$= 2.93 + 1.41 + 3.75 + 1.15 = 9.24 \quad [\chi^2_{.05(1)} = 3.84]$$

Comment: Reject H0, There is a relationship between results and job satisfaction.